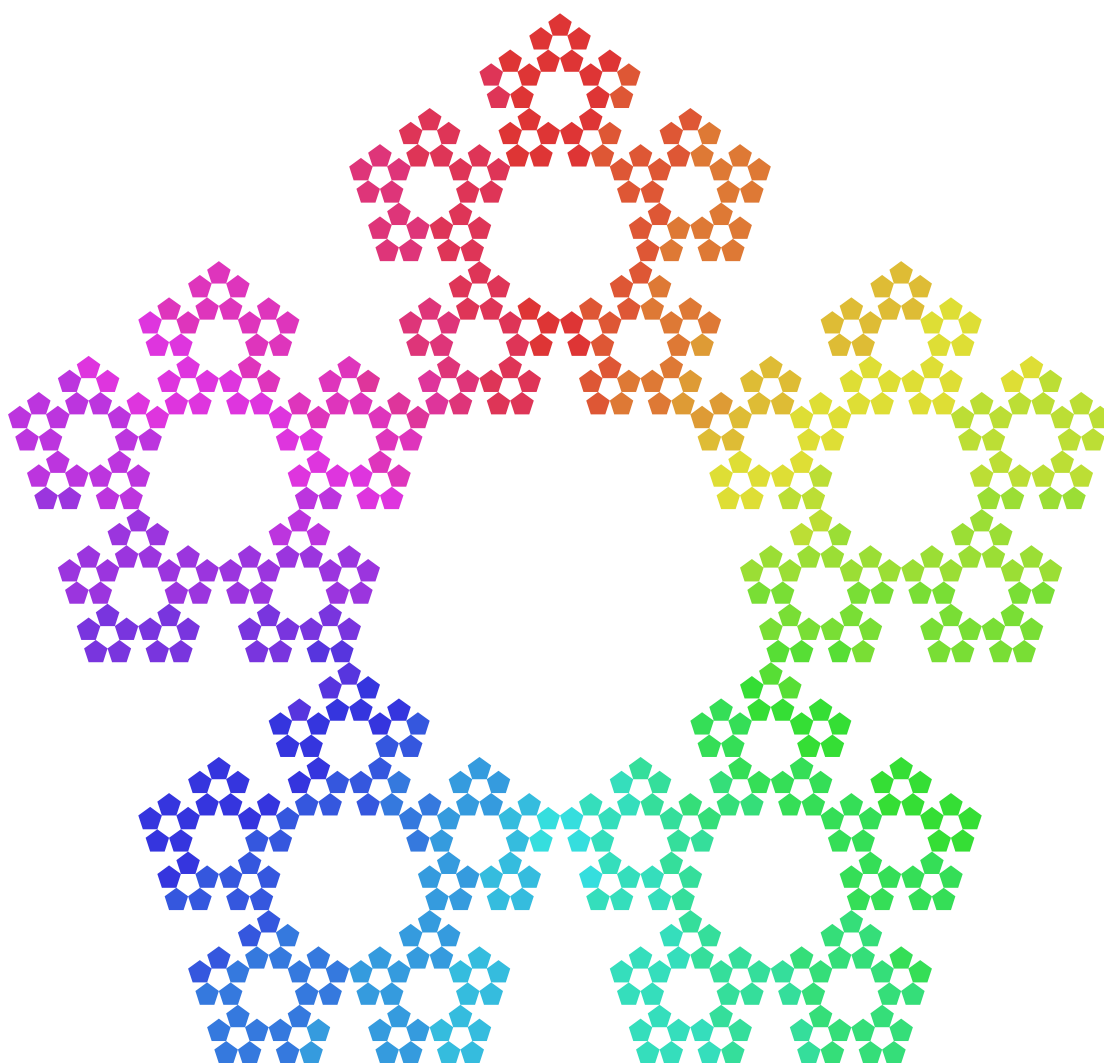


YEAR 7 MATHEMATICS

Chapter 1

Integers

LEARN IT • PRACTISE IT • REMEMBER IT



1.01 Integers and opposites

1.02 The number line

1.03 Comparing and ordering integers

1.04 Adding integers

1.05 Subtracting integers

1.06 Multiplying integers

1.07 Dividing integers

1.08 Order of operations with integers

1.09 Integers on a calculator

1.10 Integer problems

R Chapter review

On the cover: a Sierpinski pentagon. It is made of 5 copies of itself, each about 0.38 times as big, again and again forever. Each lesson in this booklet has its own colour from the rainbow around it.

Every lesson is **one double-sided sheet** in its own colour of the rainbow. A coloured band and an edge tab show where each new exercise starts. **Page 1:** learn it. **Page 2:** practise it.

WORKED EXAMPLE

Shaded, with the working done for you in blue. **Read it** and ask yourself why each step works.



YOUR TURN

White, with a strong border. **You write here.** Each one gives you a little less help than the last.

- Do now** Start straight away, from memory: quick questions from earlier lessons.
- Think first** Have a go at a puzzle before the lesson. It is fine to be unsure.
- Words and Big idea** The new words and the one idea that today is all about.
- Check** Everyone answers one question. It tells your teacher who is ready for page 2.
- Spot the pattern** Each question changes one thing. Look for what happens to the answer.
- Practice** Drill today's skill until it is quick, then mix it up with a few earlier ideas.
- Go further** Finished? Puzzles that make you think harder.
- Exit ticket** Three questions on your own. Then circle how you went.

PAGE 1: LEARN IT

Exercise band

Do now

Think first

Words - Big idea

Worked example

Your turn

Check

PAGE 2: PRACTISE IT

Lesson band

Spot the pattern

Practice drill, then mix it up

Go further

Exit ticket

Contents and progress tracker

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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the pentagon

The Sierpinski pentagon is a **fractal**: zoom in on any corner and you see the whole shape again. Every time you go one level deeper, there are 5 times as many pentagons.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

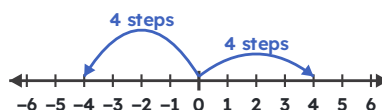
Integers and opposites

Today I will use integers and find opposites.

NAME _____

CLASS _____

DATE _____

⚡ DO NOW From memory. No notes.**1** Order smallest to largest: 12, 3, 30, 0**2** What is $8 - 8$?**3** It is 0°C . It gets 4 degrees colder. What is the temperature?**4** What number is halfway between 0 and 10?**💡 THINK FIRST** A lift is at the ground floor, level 0. It goes down 2 floors to the car park. Which level is that? Have a go. It's fine to be unsure.**📖 WORDS****integer:** a whole number: ... -2, -1, 0, 1, 2 ...**opposite:** same distance from 0, other side. $-(-5)$: say "the opposite of negative five".**BIG IDEA****Negative numbers are left of 0. A number and its opposite are the same distance from 0, on opposite sides.****📖 LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**🔍 WORKED EXAMPLE 1**

Write an integer for 5 m below sea level.

below 0: negative

 $= -5$

? Why? Why is "below sea level" negative?

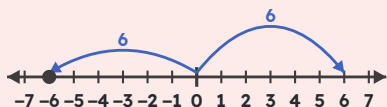
**✎ YOUR TURN 1**

Write an integer for a loss of \$8.

a loss: _____

 $=$ _____**🔍 WORKED EXAMPLE 2**

The opposite of -6

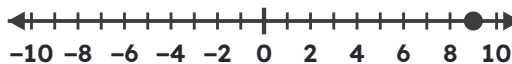
 -6 is 6 steps left of 0

opposite: 6

? Why? What is the opposite of 0? Why?

**✎ YOUR TURN 2**

The opposite of 9



___ is ___ steps right of ___

🔍 WORKED EXAMPLE 3 $-(-7)$ $=$ the opposite of -7 $= 7$? Why? Read $-(-7)$ aloud, using the word "opposite".**✎ YOUR TURN 3** $-(-11)$ **🎯 CHECK** Everyone answers. Circle one letter, then hold it up.

What is the opposite of -8?

A -8

B 8

C 0

D -7

**SPOT THE PATTERN** Find each one. Each question changes one thing.

- 1 opposite of 3 = _____ 2 opposite of 2 = _____ 3 opposite of 1 = _____ 4 opposite of 0 = _____
- 5 opposite of -1 = _____ 6 opposite of -2 = _____ 7 $-(-2)$ = _____ 8 $-(-20)$ = _____

? What do you notice? The number goes down by 1 each time. What happens to its opposite?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: USE INTEGERS AND FIND OPPOSITES**

- 1 opposite of 7 _____ 2 opposite of -3 _____ 3 opposite of 12 _____
- 4 opposite of -25 _____ 5 opposite of 0 _____ 6 $-(-4)$ _____
- 7 $-(-9)$ _____ 8 $-(-15)$ _____ 9 $-(-1)$ _____
- 10 6 m below sea level _____ 11 a gain of \$10 _____ 12 3 floors down _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 A diver is 20 m below sea level. Write this as an integer.
- 14 Which is further from 0: -7 or 5?

- 15 Which two integers are 10 steps from 0?

- 16 List the integers between -3 and 2.

from Year 6

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Mia says "the opposite of a number is always negative". Is she right? Explain.

- 2 Two opposites are 18 steps apart. What are they?

- 3 Which integers from -5 to 5 are more than 3 steps from 0?

**EXIT TICKET** On your own. Hand it in.

- 1 Write an integer for 4°C below zero.

- 2 What is the opposite of -13?

- 3 $-(-2)$

Today I can use integers and find opposites.

Not yet

Nearly

Got it

The number line

Today I will read number lines and find distances between integers.

DO NOW From memory. No notes.

1 What is the opposite of -4 ?

1.01

2 $-(-10)$

1.01

3 Write an integer for 3 floors below ground.

1.01

4 Count by 5s: 15, 20, 25, ... What comes next?

THINK FIRST A dot is halfway between -4 and 0 . What number is it at?

Have a go. It's fine to be unsure.

WORDS

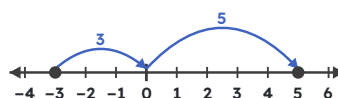
scale: how much each gap is worth.

distance: how many steps apart. Never negative.

halfway: the same distance from both ends.

BIG IDEA

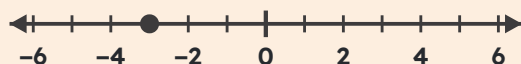
Numbers get larger to the right. To find a distance, count the gaps. Across 0? Count to 0, then on from 0.



LEARN IT Read the worked example. Then do your turn beside it. Stuck? Look back at the example.

WORKED EXAMPLE 1

Which number is at the dot?



one gap left of -2

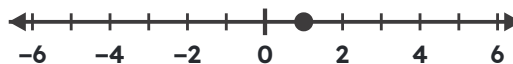
$= -3$

? Why? Only every second mark has a number. How did we still find it?



YOUR TURN 1

Which number is at the dot?

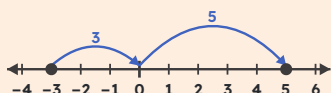


one gap right of ____

$=$ ____

WORKED EXAMPLE 2

How far is it from -3 to 5 ?



3 to 0, 5 from 0

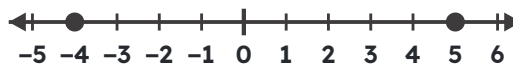
$3 + 5 = 8$

? Why? Why do we add the two parts?



YOUR TURN 2

How far is it from -4 to 5 ?



____ to ____, ____ from ____

WORKED EXAMPLE 3

Which integer is halfway between -8 and 2 ?

10 apart, half is 5

5 right of -8 : -3

? Why? How can you check that -3 is halfway?



YOUR TURN 3

Which integer is halfway between -6 and 4 ?

CHECK Everyone answers. Circle one letter, then hold it up.

How far apart are -4 and 3 ?

(A) 1 (B) 7 (C) -7 (D) -1

**SPOT THE PATTERN** How far apart are the two numbers? Each question moves one end.

- 1 -1 to $3 =$ _____ 2 -2 to $3 =$ _____ 3 -3 to $3 =$ _____ 4 -4 to $3 =$ _____
 5 -4 to $2 =$ _____ 6 -4 to $1 =$ _____ 7 -4 to $0 =$ _____ 8 -4 to $-1 =$ _____

? What do you notice? What happens to the distance when one end moves 1 step outwards? 1 step inwards?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: READ NUMBER LINES AND FIND DISTANCES BETWEEN INTEGERS**

- 1 distance -2 to 3 _____ 2 distance -5 to 1 _____ 3 distance -4 to -1 _____
 4 distance -8 to 7 _____ 5 start -3 , move 5 right _____ 6 start 4 , move 9 left _____
 7 start -6 , move 2 left _____ 8 start -10 , move 13 right _____ 9 halfway: -4 and 2 _____
 10 halfway: -10 and 0 _____ 11 halfway: -9 and -1 _____ 12 halfway: -7 and 5 _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 A number line goes $-20, -10, 0, \dots$ What are the next two numbers? 14 Point A is at -6 . Point B is 10 to the right of A. Where is B?

- 15 A lift goes from level -3 to level 6. How many floors is that?

16 $-(-3)$

from 1.01

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Two integers are 12 apart. Halfway between them is -2 . What are they?

 2 Start at -5 . Move 3 right, then 8 left, then 6 right. Where do you land? How many steps did you walk?

 3 How many integers are between -6 and 4 , not counting -6 and 4 ?

**EXIT TICKET** On your own. Hand it in.

- 1 How far is it from -7 to 2 ? 2 Which integer is halfway between -10 and 4 ? 3 Start at 3 and move 8 left. Where do you land?

Today I can read number lines and find distances between integers.

Not yet

Nearly

Got it

Comparing and ordering integers

Today I will compare and order integers.

NAME _____

CLASS _____

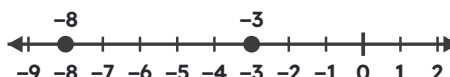
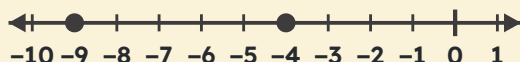
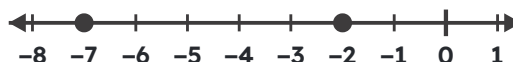
DATE _____

⚡ DO NOW From memory. No notes.**1** How far is it from **1.02** -5 to 3 ?**2** Start at -1 and **1.02** move 6 left. Where do you land?**3** What is the **1.01** opposite of -15 ?**4** Which integer is **1.02** halfway between -6 and 2 ?**💡 THINK FIRST** Which is colder: -8°C or -3°C ? Have a go. It's fine to be unsure.**📖 WORDS**

< means "is less than".

> means "is greater than".

ascending: smallest to largest.

BIG IDEAFurther right means larger. So $-3 > -8$, even though 8 is more than 3 .**📖 LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**👁 WORKED EXAMPLE 1**Which is larger: -9 or -4 ? -4 is right of -9 larger: -4 **? Why?** Sam says -9 is larger because 9 is more than 4 . What would you tell Sam?**✎ YOUR TURN 1**Which is larger: -2 or -7 ?___ is right of ___larger: ___**👁 WORKED EXAMPLE 2**Write < or >: -6 ☐ -1 -6 is left of -1 $-6 < -1$ **? Why?** How do you remember which way < and > point?**✎ YOUR TURN 2**Write < or >: 0 ☐ -5 ___ is right of ___**👁 WORKED EXAMPLE 3**Order smallest to largest: $3, -5, 0, -2$ smallest: -5 $-5, -2, 0, 3$ **? Why?** Where would your answer sit on a number line?**✎ YOUR TURN 3**Order smallest to largest: $-1, 4, -6, 2$ **🎯 CHECK** Everyone answers. Circle one letter, then hold it up.

Which is the smallest?

(A) -2 (B) 0 (C) -9 (D) -5

SPOT THE PATTERN Write $<$ or $>$. Each question changes one thing.

- 1 $3 \square 5$ _____ 2 $-3 \square 5$ _____ 3 $-3 \square -5$ _____ 4 $3 \square -5$ _____
 5 $-5 \square -3$ _____ 6 $-30 \square -5$ _____ 7 $-5 \square 0$ _____ 8 $0 \square -30$ _____

? What do you notice? Compare 1 and 3. Putting a $-$ on both numbers did what to the sign?

PRACTICE Get today's skill quick and accurate first. Then mix it up.

DRILL: COMPARE AND ORDER INTEGERS

- 1 $-4 \square 2$ _____ 2 $-6 \square -1$ _____ 3 $0 \square -3$ _____
 4 $-8 \square -9$ _____ 5 $5 \square -5$ _____ 6 $-12 \square -2$ _____
 7 $-20 \square -19$ _____ 8 $-1 \square -10$ _____ 9 smaller: -7 or -3 _____
 10 larger: -15 or -51 _____ 11 smallest first: $3, -2, -5$ _____ 12 smallest first: $-1, -8, 0,$
 -4 _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 Which is colder: -15°C or -12°C ? 14 Write an integer that is between -5 and -3 .

- 15 Order largest to smallest: $-4, 7, -10, 0$

- 16 How far is it from -11 to -2 ?

from 1.02

GO FURTHER Finished? Try these. Explain your thinking.

- 1 Always, sometimes or never? "A negative number is smaller than a positive number." Explain.

- 2 I am an odd integer. I am less than -2 and greater than -6 . What could I be?

- 3 Order these coldest first: $-7^{\circ}\text{C}, 4^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}$. How far apart are the coldest and warmest?

EXIT TICKET On your own. Hand it in.

- 1 Write $<$ or $>$: $-8 \square -3$ 2 Order smallest to largest: $2, -9, -1, -4$ 3 Which is warmer: -6°C or -10°C ?

Today I can compare and order integers.

Not yet

Nearly

Got it

Adding integers

Today I will add positive and negative integers.

NAME _____

CLASS _____

DATE _____

⚡ DO NOW From memory. No notes.**1** $-7 \square -2$ Write $<$ or $>$.

1.03

2 Order smallest to largest: 4, -6, 0, -1

1.03

3 What is the opposite of -9?

1.01

4 Start at -4. Move 6 right. Where do you land?

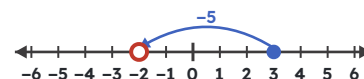
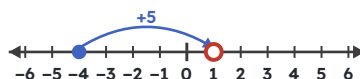
1.02

💡 THINK FIRST It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon? Have a go. It's fine to be unsure.**📖 WORDS**

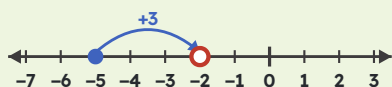
sum: the answer to an addition.

-3: say "negative three".

zero pair: +1 and -1 make 0.

BIG IDEAStart at the first number. Adding a positive: move right. Adding a negative: move left.**📖 LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**👁 WORKED EXAMPLE 1**

$-5 + 3$



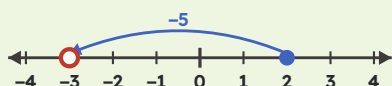
start at -5, move 3 right

$= -2$

? Why? We moved right. Why?

👁 WORKED EXAMPLE 2

$2 + (-5)$



start at 2, move 5 left

$= -3$

? Why? $2 - 5$ also moves 5 left from 2. What does that tell you about $2 + (-5)$?**👁 WORKED EXAMPLE 3**

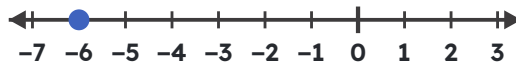
$-4 + (-3)$

$= -4 - 3$

$= -7$

? Why? Tom says $-4 + (-3) = 7$. What went wrong?**✎ YOUR TURN 1**

$-6 + 4$

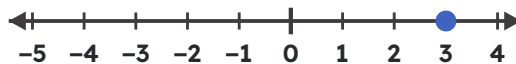


start at ____, move ____ right

$= \underline{\hspace{1cm}}$

✎ YOUR TURN 2

$3 + (-7)$



start at ____, move ____ left

$= \underline{\hspace{1cm}}$

✎ YOUR TURN 3

$-2 + (-6)$

$= \underline{\hspace{1cm}}$

🎯 CHECK Everyone answers. Circle one letter, then hold it up.Work out $-2 + 5$

- (A) -7 (B) 3 (C) -3 (D) 7

**SPOT THE PATTERN** Each question changes one thing. Work them out, then look down the answers.

$$1 \quad -3 + 3 = \underline{\hspace{2cm}} \quad 2 \quad -3 + 2 = \underline{\hspace{2cm}} \quad 3 \quad -3 + 1 = \underline{\hspace{2cm}} \quad 4 \quad -3 + 0 = \underline{\hspace{2cm}}$$

$$5 \quad -3 + (-1) = \underline{\hspace{2cm}} \quad 6 \quad -3 + (-2) = \underline{\hspace{2cm}} \quad 7 \quad -3 + (-3) = \underline{\hspace{2cm}} \quad 8 \quad -30 + (-30) = \underline{\hspace{2cm}}$$

? What do you notice? What happens to the answer each time the number added goes down by 1?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: ADD POSITIVE AND NEGATIVE INTEGERS**

$$1 \quad -5 + 2 \underline{\hspace{2cm}} \quad 2 \quad -3 + 7 \underline{\hspace{2cm}} \quad 3 \quad -8 + 8 \underline{\hspace{2cm}}$$

$$4 \quad -1 + 6 \underline{\hspace{2cm}} \quad 5 \quad 4 + (-6) \underline{\hspace{2cm}} \quad 6 \quad 2 + (-9) \underline{\hspace{2cm}}$$

$$7 \quad -3 + (-4) \underline{\hspace{2cm}} \quad 8 \quad -7 + (-2) \underline{\hspace{2cm}} \quad 9 \quad -12 + 5 \underline{\hspace{2cm}}$$

$$10 \quad 15 + (-20) \underline{\hspace{2cm}} \quad 11 \quad -9 + (-9) \underline{\hspace{2cm}} \quad 12 \quad -30 + 45 \underline{\hspace{2cm}}$$

MIX IT UP: READ EACH ONE CAREFULLY

13 A diver is at -12 m. She rises 5 m. Where is she now?

$$14 \quad -6 + \square = -1$$

15 It is -9°C . It rises 4°C , then 7°C more. What is the temperature now?

16 How far apart are -8 and 5 ?

from 1.02

**GO FURTHER** Finished? Try these. Explain your thinking.

1 Always, sometimes or never true? "Adding a negative number makes the answer smaller." Explain.

2 Find two integers that add to -3 . Then find two more, where one of them is positive.

3 Start at -4 . Add 2, 6 and -5 in any order. Where do you finish? Does the order matter?

**EXIT TICKET** On your own. Hand it in.

$$1 \quad -7 + 3$$

$$2 \quad 4 + (-9)$$

3 It is -6°C . It warms up 10°C . What is the temperature now?

Today I can add positive and negative integers.

Not yet

Nearly

Got it

Subtracting integers

Today I will subtract positive and negative integers.

NAME _____

CLASS _____

DATE _____

DO NOW From memory. No notes.

1 $-7 + 4$

1.04

2 $5 + (-8)$

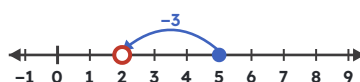
1.04

3 Write $<$ or $>$: -6 ☐ -2

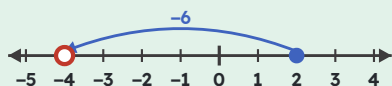
1.03

4 What is the opposite of -5 ?

1.01

THINK FIRST It is 3°C . It gets 5 degrees colder. What is the temperature now? Have a go. It's fine to be unsure.**WORDS****difference:** the answer to a subtraction. $5 - (-3)$: say "five subtract negative three".**opposite:** swap the sign: 3 and -3 .**BIG IDEA****Subtracting is adding the opposite.** $5 - 3 = 5 + (-3)$: move left. $5 - (-3) = 5 + 3$: move right.**LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**WORKED EXAMPLE 1**

$2 - 6$

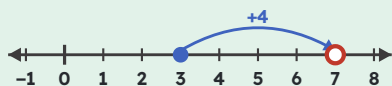


start at 2, move 6 left

$= -4$

? Why? Why is $2 - 6$ negative?**WORKED EXAMPLE 2**

$3 - (-4)$



$= 3 + 4$

$= 7$

? Why? Why does subtracting a negative move right?

WORKED EXAMPLE 3

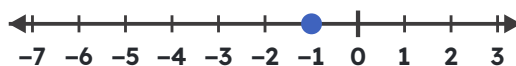
$-6 - (-9)$

$= -6 + 9$

$= 3$

? Why? Ali says $3 - (-2) = 1$. What went wrong?**YOUR TURN 1**

$-1 - 4$

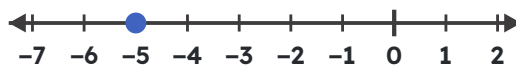


start at ____, move ____ left

$= \underline{\hspace{1cm}}$

YOUR TURN 2

$-5 - (-2)$



$= \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$

YOUR TURN 3

$-8 - (-3)$

CHECK Everyone answers. Circle one letter, then hold it up.Work out $4 - (-3)$

(A) 1

(B) 7

(C) -7

(D) -1

**SPOT THE PATTERN** Each question subtracts 1 less than the one before.

- 1 $5 - 3 =$ _____ 2 $5 - 2 =$ _____ 3 $5 - 1 =$ _____ 4 $5 - 0 =$ _____
- 5 $5 - (-1) =$ _____ 6 $5 - (-2) =$ _____ 7 $5 - (-3) =$ _____ 8 $-5 - (-3) =$ _____

? **What do you notice?** The number subtracted goes down by 1. What happens to the answer? What does that say about subtracting a negative?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: SUBTRACT POSITIVE AND NEGATIVE INTEGERS**

- 1 $3 - 7$ _____ 2 $0 - 5$ _____ 3 $-2 - 4$ _____
- 4 $-6 - 6$ _____ 5 $5 - (-2)$ _____ 6 $-3 - (-8)$ _____
- 7 $-4 - (-4)$ _____ 8 $-9 - (-2)$ _____ 9 $1 - 10$ _____
- 10 $-12 - 8$ _____ 11 $7 - (-7)$ _____ 12 $-15 - (-20)$ _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 It is -4°C . It gets 7°C colder. What is the temperature now?
- 14 How much warmer is 6°C than -2°C ? (warmer – colder)

15 $10 - (-5) - 20$

16 $-3 + (-6)$

from 1.04

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Is $7 - (-2)$ the same as $7 + 2$? Use a number line to explain.

- 2 Find a positive integer and a second integer whose difference (first – second) is -5 .

- 3 A submarine is at -120 m. A plane is at 350 m. How far apart are they?

**EXIT TICKET** On your own. Hand it in.

1 $-2 - 7$

2 $6 - (-5)$

3 $-4 - (-9)$

Today I can subtract positive and negative integers.

Not yet

Nearly

Got it

Multiplying integers

Today I will multiply integers and choose the sign.

NAME _____

CLASS _____

DATE _____

DO NOW From memory. No notes.

1 $-3 - 5$

1.05

2 $2 - (-6)$

1.05

3 $(-4) + (-4) + (-4)$

1.04

4 7×8

Year 6

THINK FIRST You lose \$3 a day for 4 days. By how much has your money changed? Have a go. It's fine to be unsure.

WORDS

product: the answer to a multiplication.

same signs: both + or both -.

different signs: one + and one -.

BIG IDEA

Multiply the numbers, then choose the sign. **Same signs:** positive. **Different signs:** negative.



LEARN IT Read the worked example. Then do your turn beside it. Stuck? Look back at the example.

WORKED EXAMPLE 1

$3 \times (-2)$



$(-2) + (-2) + (-2)$

$= -6$

? Why? Why is $3 \times (-2)$ negative?



YOUR TURN 1

$4 \times (-3)$



$(\quad) + (\quad) + (\quad) + (\quad)$

$= \quad$

WORKED EXAMPLE 2

-5×4

different signs: -

$5 \times 4 = 20, \text{ so } -20$

? Why? Is -5×4 the same as $4 \times (-5)$? Why?



YOUR TURN 2

$6 \times (-7)$

different signs: _____

WORKED EXAMPLE 3

$-3 \times (-6)$

same signs: +

$3 \times 6 = 18, \text{ so } 18$

? Why? Page 2 has a pattern that shows why $- \times -$ is positive. What does it show?



YOUR TURN 3

$-8 \times (-5)$

CHECK Everyone answers. Circle one letter, then hold it up.

Work out $-4 \times (-5)$

- (A) -20 (B) 20 (C) -9 (D) 9

**SPOT THE PATTERN** The first number goes down by 1 each time.

1 $3 \times (-4) =$ _____ 2 $2 \times (-4) =$ _____ 3 $1 \times (-4) =$ _____ 4 $0 \times (-4) =$ _____

5 $-1 \times (-4) =$ _____ 6 $-2 \times (-4) =$ _____ 7 $-3 \times (-4) =$ _____ 8 $-3 \times 4 =$ _____

? What do you notice? What happens to the answer each time? What does that show about $- \times -$?**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: MULTIPLY INTEGERS AND CHOOSE THE SIGN**

1 $4 \times (-3)$ _____ 2 -5×2 _____ 3 $-6 \times (-2)$ _____

4 $7 \times (-4)$ _____ 5 $-3 \times (-9)$ _____ 6 -8×5 _____

7 $-7 \times (-7)$ _____ 8 $9 \times (-6)$ _____ 9 $-10 \times (-4)$ _____

10 -12×3 _____ 11 $-2 \times 3 \times (-4)$ _____ 12 $-2 \times (-2) \times (-2)$ _____

MIX IT UP: READ EACH ONE CAREFULLY

13 A diver goes down 5 m every minute for 6 minutes. What is her change in height?

14 $\square \times (-4) = 20$

15 Which is larger: -2×6 or $-2 \times (-6)$?

16 $-7 + (-3)$

from 1.04

**GO FURTHER** Finished? Try these. Explain your thinking.1 Find all the pairs of integers that multiply to -12 .2 Always, sometimes or never? "Negative $\times$ negative $\times$ negative is negative." Explain.3 Work out $(-1) \times (-1) \times (-1) \times (-1) \times (-1)$. What if there were 100 of them?**EXIT TICKET** On your own. Hand it in.

1 -6×4

2 $-9 \times (-2)$

3 The temperature falls 3°C each hour for 5 hours. What is the total change?

Today I can multiply integers and choose the sign.

Not yet

Nearly

Got it

Dividing integers

Today I will divide integers and choose the sign.

NAME _____

CLASS _____

DATE _____

DO NOW From memory. No notes.

1 -6×7

1.06

2 $-4 \times (-8)$

1.06

3 $-2 - 9$

1.05

4 Order smallest to largest: $-4, 2, -8$

1.03

THINK FIRST A debt of \$24 is shared equally by 4 people. How much does each person owe? Have a go. It's fine to be unsure.**WORDS**

quotient: the answer to a division.

fact family: $3 \times (-4) = -12$, so $-12 \div 3 = -4$.

share: split into equal groups.

BIG IDEA

Division undoes multiplication, so the sign rules are the same.

Same signs: positive. Different signs: negative.**LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**WORKED EXAMPLE 1**

$-12 \div 3$



$3 \times (-4) = -12$

so $-12 \div 3 = -4$

? Why? How does the $\times$ fact help with the $\div$?**WORKED EXAMPLE 2**

$28 \div (-4)$

different signs: -

$28 \div 4 = 7$, so -7

? Why? How can you check that $28 \div (-4) = -7$?**WORKED EXAMPLE 3**

$-30 \div (-6)$

same signs: +

$30 \div 6 = 5$, so 5

? Why? Jo says $-30 \div (-6) = -5$. What went wrong?**YOUR TURN 1**

$-15 \div 5$



___ $\times$ (___) = ___

so ___ $\div$ ___ = ___

YOUR TURN 2

$-36 \div 9$

different signs: _____

YOUR TURN 3

$-48 \div (-8)$

CHECK Everyone answers. Circle one letter, then hold it up.Work out $-20 \div (-4)$

(A) -5

(B) 5

(C) -16

(D) 80

**SPOT THE PATTERN** Each question changes one sign or one number.

- 1 $12 \div 3 =$ _____ 2 $-12 \div 3 =$ _____ 3 $12 \div (-3) =$ _____ 4 $-12 \div (-3) =$ _____
- 5 $-12 \div (-4) =$ _____ 6 $-12 \div (-6) =$ _____ 7 $-12 \div 6 =$ _____ 8 $-120 \div (-6) =$ _____

? What do you notice? Which answers are negative? What do their questions have in common?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: DIVIDE INTEGERS AND CHOOSE THE SIGN**

- 1 $-12 \div 3$ _____ 2 $-20 \div 5$ _____ 3 $18 \div (-6)$ _____
- 4 $-24 \div (-8)$ _____ 5 $-35 \div 7$ _____ 6 $42 \div (-7)$ _____
- 7 $-36 \div (-4)$ _____ 8 $-50 \div 10$ _____ 9 $-64 \div (-8)$ _____
- 10 $81 \div (-9)$ _____ 11 $-100 \div (-25)$ _____ 12 $-60 \div 4 \div (-3)$ _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 $\square \div 4 = -5$
- 14 The temperature fell 18°C in 6 hours, the same each hour. What was the change each hour?

15 $-80 \div 4 \div (-2)$

16 -6×7

from 1.06

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Find the average of -6 , 2 , -5 and -3 . (Add them, then divide by 4.)

- 2 Write three different divisions that have the answer -4 .

- 3 Always, sometimes or never? "Dividing makes a number smaller." Explain.

**EXIT TICKET** On your own. Hand it in.

- 1 $-35 \div 5$
- 2 $-72 \div (-8)$
- 3 A debt of \$60 is shared by 3 people. Write each share as an integer.

Today I can divide integers and choose the sign.

Not yet

Nearly

Got it

Order of operations with integers

Today I will use the order of operations with integers.

NAME _____

CLASS _____

DATE _____

DO NOW From memory. No notes.

1 $-18 \div 3$

1.07

2 $-4 \times (-6)$

1.06

3 $5 - 9$

1.05

4 $-8 + 3$

1.04

THINK FIRST Two students work out $2 + 3 \times 4$. One gets 20. One gets 14. Who is right? Have a go. It's fine to be unsure.**WORDS**order of operations: brackets, then $\times$ and $\div$, then $+$ and $-$.

left to right: for a tie, work from left to right.

evaluate: work out the value.

BIG IDEA1. Brackets. 2. $\times$ and $\div$, left to right. 3. $+$ and $-$, left to right. The integer rules still work at every step.**LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**WORKED EXAMPLE 1**

$-3 + 4 \times 2$

$= -3 + 8$

$= 5$

? Why? Why do we do 4×2 first?**YOUR TURN 1**

$-10 + 3 \times 2$

$= ___ + ___$

$= ___$

WORKED EXAMPLE 2

$(-3 + 7) \times 2$

$= 4 \times 2$

$= 8$

? Why? How did the brackets change the answer?

**YOUR TURN 2**

$(1 - 6) \times 3$

$= ___ \times ___$

WORKED EXAMPLE 3

$6 - 2 \times (-3)$

$= 6 - (-6)$

$= 6 + 6$

$= 12$

? Why? Mia says $6 - 2 \times (-3) = -12$. What went wrong?**YOUR TURN 3**

$-5 - 4 \times (-2)$

CHECK Everyone answers. Circle one letter, then hold it up.Work out $10 - 3 \times (-2)$

(A) -14

(B) 4

(C) 16

(D) -16

SPOT THE PATTERN The same numbers, with one change each time.

- 1 $2 + 3 \times 4 =$ _____ 2 $2 - 3 \times 4 =$ _____ 3 $-2 - 3 \times 4 =$ _____ 4 $-2 - 3 \times (-4) =$ _____
 5 $(-2 - 3) \times (-4) =$ _____ 6 $(-2 + 3) \times (-4) =$ _____ 7 $-2 + 3 \times (-4) =$ _____ 8 $-20 + 30 \times (-4) =$ _____

? What do you notice? Find two questions with the same numbers but different answers. What made them different?

PRACTICE Get today's skill quick and accurate first. Then mix it up.

DRILL: USE THE ORDER OF OPERATIONS WITH INTEGERS

- 1 $-4 + 2 \times 3$ _____ 2 $5 - 3 \times 4$ _____ 3 $-2 \times 3 + 1$ _____
 4 $(-2 + 5) \times 2$ _____ 5 $(1 - 4) \times 3$ _____ 6 $-12 \div 3 + 6$ _____
 7 $8 - 10 \div 2$ _____ 8 $-3 \times (-2) - 7$ _____ 9 $(-6 - 2) \div 4$ _____
 10 $-10 + 4 \times (-2)$ _____ 11 $6 - 2 \times (-5)$ _____ 12 $-20 \div 5 \times 2$ _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 $-2 \times 3 + 4 \times (-2)$ _____ 14 $(-9 + 3) \div (-2)$ _____

15 $-3 \times (2 - 6)$ _____

16 $-12 \div (-4)$ _____

from 1.07

GO FURTHER Finished? Try these. Explain your thinking.

- 1 Put brackets in to make it true: $-4 + 6 \times 3 = 6$

- 2 Use -2 , 3 and 5 once each, with $+$ and $\times$, and brackets if you like. What is the smallest answer you can make?

- 3 Write a calculation with brackets, $\times$ and $-$ that has the answer -12 .

EXIT TICKET On your own. Hand it in.

- 1 $-6 + 2 \times 4$ _____ 2 $(-6 + 2) \times 4$ _____ 3 $9 - 3 \times (-2)$ _____

Today I can use the order of operations with integers.

Not yet

Nearly

Got it

Integers on a calculator

Today I will use a calculator with integers and check answers.

DO NOW From memory. No notes.

1 $-3 + 2 \times 5$

1.08

2 $(4 - 9) \times 2$

1.08

3 $-56 \div 8$

1.07

4 $-7 \times (-6)$

1.06

THINK FIRST Without a calculator: is -48×21 about -100 , about $-1\ 000$, or about $-10\ 000$? Have a go. It's fine to be unsure.

WORDS

(-) key: makes a number negative.

- key: subtracts.

estimate: a quick answer from rounded numbers.

BIG IDEA

Use **(-)** for a negative number, **-** to subtract. Got a **+/-** key? Press it after the number. Estimate first, then check.

(-) **4** **8** **x** **2** **1** **=**

LEARN IT Read the worked example. Then do your turn beside it. Stuck? Look back at the example.

WORKED EXAMPLE 1

$-37 + 85$

keys: **(-)** **37** **+** **85** **=**

$= 48$

? **Why?** What could go wrong if you pressed **-** instead of **(-)** at the start?

→

YOUR TURN 1

$-126 + 59$

keys: **(-)** ____ **+** ____ **=**

$=$ ____

WORKED EXAMPLE 2

-48×21

estimate: $-50 \times 20 = -1\ 000$

$= -1\ 008$

? **Why?** How does the estimate help?

→

YOUR TURN 2

-39×31

estimate: ____ $\times$ ____ $=$ ____

WORKED EXAMPLE 3

$(-35 - 47) \times 12$

bracket: **-82**

$= -984$

? **Why?** Why must you type the brackets?

→

YOUR TURN 3

$(-28 + 13) \times (-6)$

CHECK Everyone answers. Circle one letter, then hold it up.

Ella presses **(-)** **6** **-** **4** **=** What does her calculator show?

(A) -2

(B) 2

(C) -10

(D) 10

**SPOT THE PATTERN** Use your calculator. Each question changes one thing.

$$1 \quad -25 \times 4 = \underline{\hspace{2cm}} \quad 2 \quad -25 \times 8 = \underline{\hspace{2cm}} \quad 3 \quad -25 \times 12 = \underline{\hspace{2cm}} \quad 4 \quad -25 \times 16 = \underline{\hspace{2cm}}$$

$$5 \quad -25 \times (-16) = \underline{\hspace{2cm}} \quad 6 \quad 25 \times (-16) = \underline{\hspace{2cm}} \quad 7 \quad -250 \times 16 = \underline{\hspace{2cm}} \quad 8 \quad -250 \times (-160) = \underline{\hspace{2cm}}$$

? What do you notice? What happens when one number gets bigger by 4? When one sign changes?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: USE A CALCULATOR WITH INTEGERS AND CHECK ANSWERS**

$$1 \quad -347 + 128 \underline{\hspace{2cm}} \quad 2 \quad -95 - 187 \underline{\hspace{2cm}} \quad 3 \quad -504 + 769 \underline{\hspace{2cm}}$$

$$4 \quad -56 \times 23 \underline{\hspace{2cm}} \quad 5 \quad 68 \times (-45) \underline{\hspace{2cm}} \quad 6 \quad -37 \times (-29) \underline{\hspace{2cm}}$$

$$7 \quad -125 \times (-24) \underline{\hspace{2cm}} \quad 8 \quad -2\,016 \div 42 \underline{\hspace{2cm}} \quad 9 \quad -1\,248 \div (-24) \underline{\hspace{2cm}}$$

$$10 \quad -4\,785 \div 87 \underline{\hspace{2cm}} \quad 11 \quad (-64 - 36) \times 15 \underline{\hspace{2cm}} \quad 12 \quad (-18 + 7) \times (-33) \underline{\hspace{2cm}}$$

MIX IT UP: READ EACH ONE CAREFULLY13 Estimate -62×19 . Then use your calculator.14 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?15 Use your calculator to find the missing number:
 $-23 \times \square = 1\,403$ 16 No calculator: $-6 + 3 \times (-2)$

from 1.08

**GO FURTHER** Finished? Try these. Explain your thinking.1 Use your calculator to find the missing number: $-37 \times \square = 1\,147$ 2 A student gets 1 008 for -48×21 . How can you tell it is wrong without working it out?3 Moscow had these temperatures: -12 , -8 , -15 , -3 and -7°C . Find the average.**EXIT TICKET** On your own. Hand it in.

1 $-234 + 97$

2 $-38 \times (-27)$

3 $(-45 - 27) \div 8$

Today I can use a calculator with integers and check answers.

Not yet

Nearly

Got it

Integer problems

Today I will solve problems with integers.

NAME _____

CLASS _____

DATE _____

⚡ DO NOW From memory. No notes.

1 $-144 \div 12$

1.07

2 $-2 \times (3 - 7)$

1.08

3 $-5 - (-8)$

1.05

4 Which is warmer: 1.03

 -14°C or -9°C ?**💡 THINK FIRST** A lift at level -2 goes up 6 floors, then down 3 floors. Which level is it on now? Have a go. It's fine to be unsure.**📖 WORDS****balance:** how much money is in an account.**overdrawn:** the balance is below \$0.**change:** new – old.**BIG IDEA**Read the story. Write a number sentence. Up, rise, earn: $\pm$. Down, fall, spend: $\pm$. A change that repeats: $\times$.**📖 LEARN IT** Read the worked example. Then do your turn beside it. Stuck? Look back at the example.**👁 WORKED EXAMPLE 1**It is -4°C at 6 am. By noon it is 9°C warmer. What is the temperature at noon?

$-4 + 9$

$= 5^{\circ}\text{C}$

? Why? How did you choose + or -?**👁 WORKED EXAMPLE 2**

Sam has \$45. Sam spends \$70. What is the balance?

$45 - 70$

$= -\$25$

? Why? What does a balance of $-\$25$ mean?**👁 WORKED EXAMPLE 3**A diver at -10 m goes down 3 m a minute for 4 minutes. Where is she now?

$4 \times (-3) = -12$

$-10 + (-12) = -22\text{ m}$

? Why? Why multiply first?**✎ YOUR TURN 1**It is 3°C at 6 pm. By midnight it is 8°C colder. What is the temperature at midnight?

$--- - ---$

$= ---^{\circ}\text{C}$

✎ YOUR TURN 2Ana has $-\$30$. She is paid \$55. What is her balance?

$--- + ---$

✎ YOUR TURN 3A diver at -40 m rises 5 m a minute for 6 minutes. Where is she now?**🎯 CHECK** Everyone answers. Circle one letter, then hold it up.A gull is at 15 m. A shark is at -20 m . How far apart are they?A 5 mB 35 mC -5 m D -35 m

**SPOT THE PATTERN** All four operations with -5 . Each changes one thing.

- 1 $-5 + 8 =$ _____ 2 $-5 + 5 =$ _____ 3 $-5 + 2 =$ _____ 4 $-5 - 2 =$ _____
- 5 $-5 - (-2) =$ _____ 6 $-5 \times 2 =$ _____ 7 $-5 \times (-2) =$ _____ 8 $-5 \div (-5) =$ _____

? What do you notice? Which two questions have the same answer? Why?

**PRACTICE** Get today's skill quick and accurate first. Then mix it up.**DRILL: SOLVE PROBLEMS WITH INTEGERS**

- 1 -5°C , then rises 8°C _____ 2 4°C , then falls 9°C _____
- 3 -2°C , then falls 6°C _____ 4 -11°C , then rises 11°C _____
- 5 level -3 , then up 5 _____ 6 level 4, then down 6 _____
- 7 have \$10, spend \$25 _____ 8 have $-\$20$, earn \$30 _____
- 9 apart: 15 m and -10 m _____ 10 apart: -4 m and -19 m _____
- 11 down 3 m a minute, 5 min _____ 12 falls 2°C an hour, 6 h _____

MIX IT UP: READ EACH ONE CAREFULLY

- 13 A quiz: $+3$ if right, -1 if wrong. Kai gets 6 right, 5 wrong. His score? _____
- 14 A balance is \$20. You spend \$35, then earn \$10. Balance? _____
- 15 A plane is at 900 m. A submarine is at -150 m. How far apart are they? _____
- 16 Order coldest first: -3°C , -13°C , 3°C , 0°C from 1.03 _____

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 In the quiz, Lea answered 10 questions and scored 14. How many were right?

- 2 It is -6°C . It warms 2°C an hour for 12 hours, then cools 3°C an hour for 4 hours. Temperature?

**EXIT TICKET** On your own. Hand it in.

- 1 It is -7°C . It rises 11°C . What is the temperature now? _____
- 2 A balance is \$15. You spend \$40. What is the balance? _____
- 3 A diver at -8 m goes down 2 m a minute for 5 minutes. Where is she? _____

Today I can solve problems with integers.

Not yet

Nearly

Got it

Chapter review

Today I will show what I remember from the whole chapter.

NAME _____

CLASS _____

DATE _____

MIXED REVIEW A No notes. Show your working. Calculator only where it says so.

1 What is the opposite of -17 ?

1.01

2 $-(-8)$

1.01

3 How far is it from -6 to 9 ?

1.02

4 Which integer is halfway between -14 and 2 ?

1.02

5 Write $<$ or $>$: -12 $\square$ -21

1.03

6 Order smallest to largest: $5, -8, 0, -3, -10$

1.03

7 $-9 + 14$

1.04

8 $3 + (-11)$

1.04

9 $-4 - 7$

1.05

10 $-2 - (-10)$

1.05

11 -7×8

1.06

12 $-6 \times (-9)$

1.06

**MIXED REVIEW B** Keep going. Each question shows which lesson it comes from.

13 $-81 \div 9$

1.07

14 $-64 \div (-8)$

1.07

15 $-5 + 3 \times (-4)$

1.08

16 $(-5 + 3) \times (-4)$

1.08

17 Calculator: $-384 \div 16$

1.09

18 Calculator: $(-73 + 28) \times 11$

1.09

19 It is -3°C . It falls 9°C . What is the temperature now?

1.10

20 A diver at -25 m rises 4 m a minute for 5 minutes. Where is she now?

1.10

21 How far apart are 40 m and -35 m ?

1.10

22 $-2 \times (-3) \times (-5)$

1.06

23 Is $6 - (-4)$ the same as $6 + 4$? Explain.

1.05

24 Put brackets in to make it true: $2 - 8 \times 3 = -18$

1.08

**WHAT NEXT?** Mark your answers with your teacher. Tick each lesson where you got every question right.

1.01

☐

1.02

☐

1.03

☐

1.04

☐

1.05

☐

1.06

☐

1.07

☐

1.08

☐

1.09

☐

1.10

☐

Not ticked? Go back to that lesson's worked examples, then try its exit ticket again.